
VANISH: Vanishing-Factor Ablation of Non-tangent Flows in Simplex and Hypercube

Roshan Bellary,¹ Audhav Durai,¹ Chang-Uk Jeong,¹ Pedram Bayat¹

¹Department of Computer and Information Science, University of Pennsylvania

Abstract

On a bounded domain, nothing in the training objective of a continuous generative model makes the learned velocity field point back inside, and samplers silently clamp the difference away. We study human DNA, protein and trinucleotide sequences on the probability simplex (four to sixty-four symbols), scored on held-out sequences, and MNIST images in the unit box. Flow matching fits held-out marginals better than matched diffusion; reward guidance steers our model toward GC-rich DNA and hydrophobic protein, at a cost in diversity. The learned field leaves the simplex on 71% of sampler steps with four symbols and on nearly all with sixty-four. Our innovation isolates the multiplicative factor of a known replicator parameterisation: alone, it removes violations at moderate step counts and improves the linear base at twenty and sixty-four symbols on every seed, while the centering term is inert. It does not reach Dirichlet or Fisher-Flow flow matching, which are feasible too. On images the factor again removes violations but not quality gaps; a mirror-map baseline is significantly better. Feasibility is cheap and every strong method has it; the factor alone buys a real but limited gain.

1 Introduction

Diffusion models reverse a stochastic corruption [1], flow matching regresses a vector field along a probability path [2], and interpolants unify the two [3]; which suits a given state space is worth settling under one protocol.

Many state spaces are *bounded*: sequences relaxed onto the simplex [4–6] live in Δ_K , images in $[0, 1]^d$. The regression target is admissible, but the fitted field is unconstrained and need not point inward. Theory for confined flows assumes it does [7]; practice repairs it by clamping, projection or thresholding [4, 8]. Sample-level violations are known [9]; how often the per-step repair fires on real data is not. The replicator parameterisation of Boll et al. [10] makes the field tangent by construction but bundles two mechanisms never separated in flow matching.

On simplex sequences (Modality 1) and MNIST in a box (Modality 2) we compare flow matching with diffusion, demonstrate guidance, ablate the replicator form, and compare with recent methods. We hypothesised that a field vanishing on the boundary would generate better, and that the gain would transfer because it depends only on boundedness.

Our contributions are summarized as follows.

- A matched comparison of flow matching and diffusion on three simplex datasets; flow matching has lower held-out marginal KL on every dataset.
- Reward guidance of the innovation model on real sequences, with its control–quality–diversity trade-off quantified.
- A measurement of how often learned fields leave the simplex on three alphabets, and a one-at-a-time ablation isolating the replicator form’s multiplicative factor.

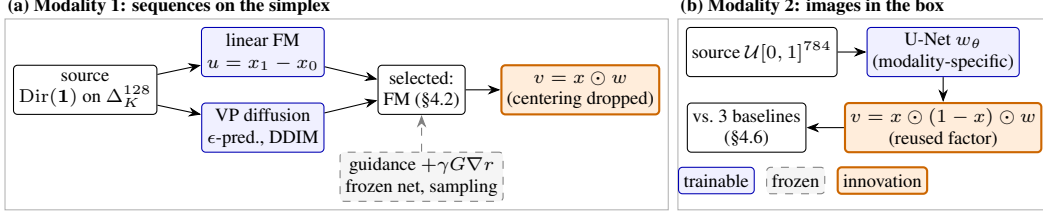


Figure 1: Study overview. Guidance acts on the frozen network; only the velocity parameterisation changes, and only the factor is carried to Modality 2. Nothing is pretrained.

- A comparison with three recent simplex methods and a no-model unigram reference.
- A transfer of the factor to $[0, 1]^{784}$ against three boundary-handling baselines: feasibility transfers, quality does not.

2 Related Work

Frameworks. Flow matching, rectified flow and SiT [2, 11, 12] and EDM, DDIM and the cosine schedule [1, 13, 14] frame our baselines.

Boundaries. Reflected flow matching reports how often unconstrained flow matching leaves the simplex and the box [9]; reflected and mirror diffusion enforce domains by reflection or a change of coordinates [15–17]. Boll et al. [10] parameterise flows with the replicator map $\text{Diag}(p) - pp^\top$; Wang and Kanamori [18] use the same face-vanishing factors on simplices and boxes for reflected score learning; Cheng et al. [19] note that such a field avoids overshooting, and PolyFlow separates fitting from integration error [20]. The parameterisation is prior art; ours is the test of the factor *without* its centering, and the measurement on real data.

Modality 1. Dirichlet flow matching follows Dirichlet paths [4]; Fisher-Flow follows Fisher–Rao geodesics on the sphere $\sqrt{p}$ [5]; Gumbel-Softmax flow matching anneals a relaxed categorical [6]. We re-implement all three on our backbone, optimiser and budget; each keeps its own path, objective and sampler (Dirichlet and Gumbel-Softmax train cross-entropy denoisers).

Modality 2. Reflection [15], mirror maps [16] and dynamic thresholding [8] are the standard ways to stay in $[0, 1]^d$.

3 Methods

3.1 Modalities, Data, and Preprocessing

Modality 1 is unconditional generation of length-128 sequences, points of Δ_K^{128} ; data are one-hot vertices. **DNA** ($K = 4$): 128-base windows of GRCh38 chr1:1–10 Mb [21]. **Codon** ($K = 64$): the same region in 384-base windows read as non-overlapping trinucleotides (3-mer tokens, not reading-frame codons). **Protein** ($K = 20$): reviewed human SwissProt entries of length 128–1000 [22], tiled into non-overlapping 128-residue windows. Held-out sets are split off by protein and by 100 kb genomic block (the same blocks for DNA and codon), deduplicated; models train on 12,000 sequences and are scored against 4,351 held-out ones. **Modality 2** is unconditional MNIST [23] generation in $[0, 1]^{784}$, mapped inside by $x \mapsto 0.98x + 0.01$; we train on 20,000 images and hold out the next 5,000 as the Fréchet reference. The simplex (a sum-constrained slice) and the box (a product of intervals) share only boundedness.

3.2 Continuous Flow Matching Baseline

With $x_1 \sim p_{\text{data}}$ and $x_0 \sim \text{Dir}(1)$ per position, $x_t = (1 - t)x_0 + tx_1$, $u_t = x_1 - x_0$, and $\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t \sim \mathcal{U}[0, 1]} \|v_\theta(x_t, t) - u_t\|_2^2$. The backbone is a dilated 1-D CNN (8 residual blocks, width 128, 664K–679K parameters); Adam at a constant 10^{-3} , batch 128, 8000 steps, final checkpoint, 8 seeds. Sampling is Euler at NFE $\in \{10, 50, 100\}$, then clamping and renormalisation.

3.3 Diffusion Baseline

On the same coordinates, $x_s = \sqrt{\bar{\alpha}_s}x_1 + \sqrt{1 - \bar{\alpha}_s}\epsilon$ with the cosine schedule [14], $\mathcal{L}_{\text{diff}}(\phi) = \mathbb{E}\|\epsilon - \epsilon_\phi(x_s, s)\|_2^2$, and deterministic DDIM [13], $x_{s'} = \sqrt{\bar{\alpha}_{s'}}\hat{x}_1 + \sqrt{1 - \bar{\alpha}_{s'}}\epsilon_\phi$ with $\hat{x}_1 = (x_s - \sqrt{1 - \bar{\alpha}_s}\epsilon_\phi)/\sqrt{\bar{\alpha}_s}$. All else matches 3.2.

3.4 Guided Generation

We steer the frozen innovation model (§3.5) toward a sequence property with the reward $r(x) = \sum_l \log \sum_{i \in S} x_{l,i}$: GC content on DNA ($S = \{C, G\}$) and hydrophobic fraction on protein ($S = \{A, I, L, M, F, V, W\}$). The drift becomes $x \odot w_\theta + \gamma \nabla r$, γ constant in t , and each Euler step is projected onto the simplex and renormalised (the Euclidean rule); nothing is retrained. A synthetic check with an exact guided law is in App. A.5.

3.5 Proposed Innovation

Hypothesis: the learned field points out of the simplex and the repairing clamp distorts samples, so a field vanishing on each face should generate better. The replicator form $v = x \odot (w - \langle x, w \rangle)$ [10] combines two mechanisms: x_i makes each component vanish on the face $x_i = 0$, and the centering gives $\sum_i v_i = 0$. We propose the factor alone,

$$v_\theta = x \odot w_\theta(x, t), \quad \mathcal{L}(\theta) = \mathbb{E}\|x_t \odot w_\theta(x_t, t) - (x_1 - x_0)\|_2^2, \quad (1)$$

with no new parameters or hyperparameters; the sum constraint is left to renormalisation. Controls: the base $v = w$, center-only $v = w - \bar{w}$ ($\bar{w}$ the coordinate mean), and the full form (Algorithm 1).

3.6 Transfer to Modality 2

The invariant idea, a factor vanishing where the domain ends, becomes $v = x \odot (1 - x) \odot w$ on the box (no sum constraint, so no centering). Only the backbone changes: a U-Net (width 48, 0.60M parameters), uniform source, Adam at 2×10^{-4} , batch 128, 6000 steps, 8 seeds, Euler at NFE $\in \{10, 50, 100\}$ with violations counted before clamping.

3.7 Model Overview

Figure 1 summarises what is trained, frozen and changed in each modality.

4 Results

4.1 Evaluation Protocol

4.1.1 Metrics

Step violation is the fraction of (sample, step) pairs whose raw Euler update has any of its $128K$ coordinates below zero, before clamping, with no tolerance. It counts steps, not mass (App. A.3), and rises with K through the coordinate count alone, so we compare it within a dataset. Quality is per-position *marginal KL* and *3-mer correlation* [24], with a unigram sampler as a floor; diversity is checked post hoc (App. A.6). Modality 2 uses Fréchet distance (FD), density and coverage [25] in the features of one classifier trained once (96.2% test accuracy).

4.1.2 Fair Comparison and Computational Budget

Within an experiment all arms share data, architecture, optimiser, steps, seeds, sample count (4,000 sequences; 5,000 images) and NFE grid; each external method keeps its own path, objective and decoding (Dirichlet and Gumbel-Softmax train cross-entropy denoisers and decode the argmax of their final posterior, as published). Seeds fix initialisation, not minibatches across objectives. Runs used A40 and A6000 GPUs, about 12.1 minutes per Modality 1 cell. All baselines are our re-implementations at our scale; none is quoted.

Table 1: Flow matching vs. diffusion on the real sequences (8 seeds): held-out marginal KL and paired differences, diffusion minus flow matching.

Dataset	FM, NFE 10	Diff., NFE 10	Diff.−FM, NFE 10	Diff.−FM, NFE 100	x_0 out (FM / Diff.)
DNA ($K=4$)	0.0061 $\pm$ 0.0043	0.7259 $\pm$ 0.4016	+0.720 [+0.447, +1.005]	+0.119 [+0.054, +0.194]	0.979 / 0.993
Protein ($K=20$)	0.3120 $\pm$ 0.0632	2.2958 $\pm$ 0.4275	+1.984 [+1.724, +2.265]	+0.301 [+0.220, +0.391]	1.000 / 1.000
Codon ($K=64$)	0.7483 $\pm$ 0.1314	1.7055 $\pm$ 0.6730	+0.957 [+0.570, +1.415]	+0.647 [+0.375, +0.991]	1.000 / 1.000

Table 2: Guiding our model (8 seeds). Hit: above the real 90th percentile; r_{top} : 3-mer r to the real top decile. Full: Table S3.

γ	DNA (GC)			Protein (hydrophobic)		
	Hit $\uparrow$	$r_{\text{top}} \uparrow$	Ham.	Hit $\uparrow$	$r_{\text{top}} \uparrow$	Ham.
0 (unguided)	0.09	0.59	0.75	0.03	0.60	0.94
0.03 (low)	0.30	0.92	0.74	0.55	0.74	0.94
0.3 (default)	1.00	0.65	0.53	1.00	0.53	0.84
3 (high)	1.00	0.55	0.44	1.00	0.50	0.72
<i>Held-out real</i>	0.09	—	0.75	0.10	—	0.94

Table 3: Decomposition (8 seeds, NFE 100). Bold: significant vs. base, or violation $< 10^{-3}$.

Arm	3-mer r			step violation		
	$K=4$	$K=20$	$K=64$	$K=4$	$K=20$	$K=64$
base w	0.894	0.753	0.127	0.713	0.967	1.000
mult $x \odot w$	0.899	0.865	0.196	0.0000	0.0000	0.0000
center $w - \bar{w}$	0.910	0.777	0.120	0.721	0.967	1.000
full replicator	0.896	0.836	0.187	0.010	0.010	0.016

4.1.3 Statistical Reporting

Mean $\pm$ s.d. over seeds and paired-bootstrap 95% intervals (20,000 resamples; significant if the interval excludes zero). At 8 seeds the bootstrap is optimistic, so headline effects also get Student- t intervals and seed counts; there is no multiplicity correction. Registered before their runs: Table 1, the held-out data and split, the corrected baselines, the Modality 2 rerun and the guidance test; not registered: the decomposition, the 3-mer endpoint and the choice of datasets.

4.2 Modality 1: Flow Matching versus Diffusion

Which family should carry the innovation? With the same backbone, flow matching attains lower held-out marginal KL at every NFE on every dataset, on at least 7 of 8 seeds (Table 1): +0.720 [+0.447, +1.005], +1.984 [+1.724, +2.265] and +0.957 [+0.570, +1.415] (diffusion minus flow matching, NFE 10), and the gap persists at NFE 100; the preregistered synthetic run agreed (Table S1). Diffusion was untuned, so we claim the ordering, not the margin; a unigram sampler reaches KL 0.002–0.019 on DNA and codon, so this metric mostly tests composition. Diffusion is also less diverse (Hamming 0.69 / 0.90 / 0.94 vs. 0.75 / 0.94 / 0.98; App. A.6).

4.3 Modality 1: Guidance Works

Can the frozen model be steered toward a property? Yes (Table 2). At the default $\gamma = 0.3$ the hit rate rises by +0.908 [+0.870, +0.933] on DNA and +0.974 [+0.965, +0.981] on protein (paired vs. unguided), with no violations or extra runtime. The cost is diversity and realism: Hamming falls by −0.215 [−0.221, −0.211] and −0.102 [−0.110, −0.096], and samples overshoot real high-property sequences; post hoc, similarity to the real top decile peaks at small strength ($\gamma = 0.1$ on DNA: 0.94 vs. 0.59 unguided).

4.4 Modality 1: Innovation Ablations

Which part of the replicator form matters? The base field leaves the simplex on 0.713, 0.967 and 1.000 of steps at $K = 4, 20, 64$ (NFE 100; Fig. S1; per coordinate it falls). In Table 3 (Fig. S3) the factor alone improves 3-mer correlation by +0.113 [+0.085, +0.139] at $K = 20$ and +0.069 [+0.049, +0.089] at $K = 64$ (8/8 and 8/8 seeds; t -intervals agree); center-only is never significant (the base target already sums to zero), and dropping the full form’s centering changes nothing (+0.009 [−0.010, +0.031]). Violations vanish at NFE ≥ 50 ; at NFE 10 8.7–10.0% remain, about one step in ten. At $K = 20$ the gain shrinks with NFE (+0.337, +0.209, +0.113), as a clamping effect would; at $K = 64$ it is not significant at NFE 10 (−0.030 [−0.065, +0.005]) and appears from NFE 50 (Fig. S2).

Table 4: Recent methods (our re-implementations), the base and a no-model reference (8 seeds, NFE 100). Violation is of each method’s own update.

Method	3-mer $r \uparrow$ (NFE 100)			Step viol. $\downarrow$
	DNA ($K=4$)	Protein ($K=20$)	Codon ($K=64$)	$K=4/20/64$
Dirichlet FM [4]	0.901 ± 0.062	0.906 ± 0.020	0.447 ± 0.020	0.00/0.00/0.00
Fisher-Flow [5]	0.909 ± 0.049	0.886 ± 0.016	0.295 ± 0.013	0.00/0.00/0.00
Gumbel-Softmax FM [6]	0.894 ± 0.052	0.705 ± 0.061	0.400 ± 0.070	0.00/0.00/0.00
Linear FM (internal base)	0.894 ± 0.088	0.753 ± 0.058	0.127 ± 0.023	0.71/0.97/1.00
Ours: $v = x \odot w$	0.899 ± 0.061	0.865 ± 0.027	0.196 ± 0.016	0.00/0.00/0.00
<i>Reference: unigram sampler (no model)</i>	-0.046 ± 0.025	0.900 ± 0.002	0.218 ± 0.002	—

Table 5: Transfer to MNIST (8 seeds, NFE 100 unless noted). All arms are flow matching in our harness: reflection and thresholding are sampler rules; mirror trains in logit space.

Method	FD $\downarrow$	Coverage $\uparrow$	Density $\uparrow$	FD@10 $\downarrow$	Step viol. $\downarrow$
Reflection sampler [15]	192.0 ± 143.5	0.624	0.478	540.3	0.1401
Mirror-map FM [16]	52.5 ± 23.9	0.685	0.480	98.9	0.0000
Dyn. thresholding on x_t [8]	195.5 ± 150.3	0.623	0.480	540.8	0.1604
Flow matching (internal base)	193.9 ± 145.6	0.623	0.478	545.3	0.1602
Ours: $v = x(1 - x) \odot w$	151.0 ± 47.4	0.410	0.302	269.6	0.0000

4.5 Modality 1: Comparison with Recent Methods

Is the method competitive? No (Table 4, Fig. S4). At $K = 4$ all are within noise; at $K = 20, 64$ Dirichlet flow matching leads us by $+0.041$ [$+0.020, +0.071$] and $+0.250$ [$+0.230, +0.269$], Fisher-Flow by $+0.020$ [$+0.001, +0.042$] and $+0.098$ [$+0.080, +0.114$]; we beat Gumbel-Softmax only at $K = 20$. All three are feasible in their own samplers (App. A.4). A no-model unigram sampler scores 0.900 and 0.218 at $K = 20, 64$: protein 3-mer correlation is nearly saturated by composition, and on codon ours (0.196) stays below the floor.

4.6 Modality 2: Transfer of the Innovation

Does the factor help in a different state space? Feasibility transfers (violation 0.0000; Table 5, Fig. S5); quality does not. The paired FD change against the base is $+42.9$ [$-41.8, +144.2$], so the preregistered claim fails; ours matches reflection and thresholding, which act as plain clamping [15], is significantly worse than mirror-map flow matching (-98.5 [$-136.7, -65.6$]), and loses coverage and density on all 8 seeds. Mirror also uses a different source (Gaussian in logit space).

4.7 Cross-Modality Analysis and Failure Modes

Does the mechanism help for the same reason in both spaces? The factor removes violations in both, but feasibility does not separate methods: every strong one is feasible (by vanishing mixtures, a change of coordinates or our factor) yet quality differs widely. Failure modes: protein barely exceeds composition; on images coverage and density fall and samples are blotchier (Fig. S6); at NFE 10 the last step overshoots; six interventions failed (Table S2). *Answer:* the factor improves the linear base at $K \geq 20$, trails recent methods, and transfers feasibility but not quality.

5 Discussion

The ablation shows causally that the factor, not the centering, is the active part; the baseline and transfer comparisons are correlational and go against us. The likeliest explanation of the gain is an inductive bias: multiplying by x_i concentrates the field where mass already is, as the Dirichlet and Gumbel mixtures also do; its decay with NFE at $K = 20$ suggests part is avoided clamping.

Limitations: small models, re-implemented baselines, post-hoc diversity checks and a protein metric near its composition floor. Supported: a vanishing factor removes violation at $\text{NFE} \geq 50$ and improves a linear flow at $K \geq 20$; its centering is unnecessary; it trails recent methods.

References

- [1] T. Karras, M. Aittala, T. Aila, and S. Laine, “Elucidating the design space of diffusion-based generative models,” in *Advances in Neural Information Processing Systems*, vol. 35, pp. 26565–26577, 2022.
- [2] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le, “Flow matching for generative modeling,” in *International Conference on Learning Representations (ICLR)*, 2023.
- [3] M. S. Albergo, N. M. Boffi, and E. Vanden-Eijnden, “Stochastic interpolants: A unifying framework for flows and diffusions,” *Journal of Machine Learning Research*, vol. 26, no. 209, pp. 1–80, 2025.
- [4] H. Stark, B. Jing, C. Wang, G. Corso, B. Berger, R. Barzilay, and T. Jaakkola, “Dirichlet flow matching with applications to DNA sequence design,” in *International Conference on Machine Learning*, 2024. arXiv:2402.05841.
- [5] O. Davis, S. Kessler, M. Petrache, İ. İ. Ceylan, M. Bronstein, and A. J. Bose, “Fisher flow matching for generative modeling over discrete data,” in *Advances in Neural Information Processing Systems*, vol. 37, pp. 139054–139084, 2024. arXiv:2405.14664.
- [6] S. Tang, Y. Zhang, A. Tong, and P. Chatterjee, “Gumbel-softmax flow matching with straight-through guidance for controllable biological sequence generation,” *arXiv preprint arXiv:2503.17361*, 2025.
- [7] R. Cont, “Reflected diffusions, no-flux continuity equations and confined lagrangian flows in bounded domains,” *arXiv preprint arXiv:2607.28344*, 2026.
- [8] C. Saharia, W. Chan, S. Saxena, L. Li, J. Whang, E. Denton, *et al.*, “Photorealistic text-to-image diffusion models with deep language understanding,” *Advances in Neural Information Processing Systems*, vol. 35, 2022. arXiv:2205.11487.
- [9] T. Xie, Y. Zhu, L. Yu, T. Yang, Z. Cheng, S. Zhang, X. Zhang, and C. Zhang, “Reflected flow matching,” in *International Conference on Machine Learning (ICML)*, 2024. arXiv:2405.16577.
- [10] B. Boll, D. Gonzalez-Alvarado, S. Petra, and C. Schnörr, “Generative assignment flows for representing and learning joint distributions of discrete data,” *arXiv preprint arXiv:2406.04527*, 2024. Revised January 2025.
- [11] X. Liu, C. Gong, and Q. Liu, “Flow straight and fast: Learning to generate and transfer data with rectified flow,” in *International Conference on Learning Representations (ICLR)*, 2023.
- [12] N. Ma, M. Goldstein, M. S. Albergo, N. M. Boffi, E. Vanden-Eijnden, and S. Xie, “SiT: Exploring flow and diffusion-based generative models with scalable interpolant transformers,” in *European Conference on Computer Vision (ECCV)*, 2024.
- [13] J. Song, C. Meng, and S. Ermon, “Denoising diffusion implicit models,” in *International Conference on Learning Representations*, 2021.
- [14] A. Q. Nichol and P. Dhariwal, “Improved denoising diffusion probabilistic models,” in *International Conference on Machine Learning*, pp. 8162–8171, 2021.
- [15] A. Lou and S. Ermon, “Reflected diffusion models,” in *International Conference on Machine Learning*, 2023. arXiv:2304.04740.
- [16] G.-H. Liu, T. Chen, E. A. Theodorou, and M. Tao, “Mirror diffusion models for constrained and watermarked generation,” in *Advances in Neural Information Processing Systems*, 2023. arXiv:2310.01236.
- [17] N. Fishman, L. Klärner, V. De Bortoli, E. Mathieu, and M. Hutchinson, “Diffusion models for constrained domains,” *Transactions on Machine Learning Research*, 2023. arXiv:2304.05364.
- [18] Z. Wang and T. Kanamori, “Should the boundary term be learned in reflected diffusion? Conormal trace and reflection masking,” *arXiv preprint arXiv:2608.03469*, 2026.

- [19] C. Cheng, J. Li, J. Peng, and G. Liu, “Categorical flow matching on statistical manifolds,” in *Advances in Neural Information Processing Systems*, 2024. arXiv:2405.16441.
- [20] J. Ma, Q. Yang, Y. Zhang, L. Yan, Z. Cao, Y. Zhang, and Y. Gao, “PolyFlow: Safe and efficient polytope-constrained flow matching with constraint embedding and projection-free update,” in *International Conference on Machine Learning*, 2026. arXiv:2606.13400.
- [21] F. J. Martin, M. R. Amode, A. Aneja, *et al.*, “Ensembl 2023,” *Nucleic Acids Research*, vol. 51, no. D1, pp. D933–D941, 2023.
- [22] The UniProt Consortium, “UniProt: the universal protein knowledgebase in 2023,” *Nucleic Acids Research*, vol. 51, no. D1, pp. D523–D531, 2023.
- [23] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, “Gradient-based learning applied to document recognition,” *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.
- [24] C. Wang, M. Uehara, Y. He, A. Wang, T. Biancalani, A. Lal, T. Jaakkola, S. Levine, H. Wang, and A. Regev, “Fine-tuning discrete diffusion models via reward optimization with applications to DNA and protein design,” in *International Conference on Learning Representations*, 2025. arXiv:2410.13643.
- [25] M. F. Naeem, S. J. Oh, Y. Uh, Y. Choi, and J. Yoo, “Reliable fidelity and diversity metrics for generative models,” in *International Conference on Machine Learning*, pp. 7176–7185, 2020.

A Appendix

Every table and figure here supports the main text: Table S1 (§4.2), Table S6 (§4.3), Figures S1, S2 and S3 (§4.4), Figure S4 (§4.5), Figures S5 and S6 (§4.6–4.7) and Table S2 (§4.7).

A.1 Dataset and Preprocessing Details

Built by `experiments/prepare_real_data_v2.py`; array hashes are in `results/data_v2_manifest.json`. DNA and codon come from one region, GRCh38 chr1:1,000,000–9,999,999 (Ensembl REST, upper-cased), cut into 128- and 384-base windows; windows containing N are dropped. 18 of 90 blocks of 100 kb are held out, the same for both, and windows straddling a train/held-out boundary are dropped. Protein uses the 16,363 reviewed human SwissProt entries of length 128–1000, tiled from the N-terminus into non-overlapping 128-residue windows (windows with non-standard residues dropped); 20% of accessions are held out, so no protein contributes to both sides. Exact duplicates are removed within each side and held-out windows identical to a training window are removed. Pools are 38,300/9,746 (protein), 56,236/13,709 (DNA) and 18,733/4,559 (codon) train/held-out windows, each permuted (seed 0) and truncated to 12,000/4,351. An earlier version kept only each protein’s first 128 residues and split nothing; it is superseded. MNIST uses the standard 60,000 training images.

A.2 Training and Sampling Details

Modality 1 CNN: kernel 5, dilation cycling 1, 2, 4, 8, 16, GroupNorm, sinusoidal time features, zero-initialised output. Modality 2 U-Net: two downsampling stages (width 48 and twice that), a mid block and symmetric upsampling with skip connections, GroupNorm, SiLU, sinusoidal time embedding per block. The mirror baseline trains flow matching on $y = \text{logit}(x)/4$ from a Gaussian source and maps back through the sigmoid, so its source differs from the other arms’. No run uses a learning-rate schedule, early stopping or checkpoint selection.

A.3 Additional Ablations

Table S1 is the preregistered §4.2 decision run. It deviated from our registration, which specified the dilated CNN and a real-data leg; the deviation is disclosed in the repository’s preregistration log, and Table 1 is the registered repair. Table S2 lists the other constraint interventions we tried; with the Modality 2 transfer, six interventions were falsified, including a soft tangency penalty, a convex-combination integrator (a no-op under uniform steps) and a $(1-t)^2$ loss reweighting that was harmful against the base ($-0.273 [-0.528, -0.049]$ at $K = 20$; the t -interval $-0.273 [-0.577, +0.031]$ includes zero, 2/8 seeds favour it). **Violation magnitude and step size.** On a synthetic $K = 8$ model the base rate is flat (0.21–0.22) over a $32\times$ step-size range and the clamped mass is 3.18×10^{-5} – 1.04×10^{-3} per step; on real DNA the rate rises from 0.24 at NFE 10. We did not measure clamped mass on real data.

Table S1: The preregistered §4.2 decision run: synthetic categorical target, small MLP, 5 seeds. Marginal KL (lower better).

Method	$K=4$, NFE 10	$K=4$, NFE 100	$K=60$, NFE 10	$K=60$, NFE 100	x_0 out, $K=4$	x_0 out, $K=60$
Flow matching	0.0232 ± 0.0096	0.0101 ± 0.0040	0.1873 ± 0.0406	0.0635 ± 0.0103	0.795	1.000
Diffusion (VP, DDIM)	0.2436 ± 0.1312	0.0414 ± 0.0424	0.2708 ± 0.0741	0.1636 ± 0.0949	0.789	1.000

A.4 Additional Baseline Results

All baseline numbers are reproduced by our group; none is quoted from the original papers. **Corrections, each preregistered before its run.** (i) Our first Fisher-Flow sampler integrated its sphere velocity in simplex coordinates; the corrected arm steps on the sphere and maps back by $p = s^2$, so its own update is feasible by construction (as a Euclidean simplex velocity its field would exit on 0.238 / 0.727 / 0.887 of steps; exponential-map steps can still leave the positive orthant of the sphere). (ii) Our first Dirichlet and Gumbel-Softmax arms regressed on velocity targets that were neither the published methods nor their own paths’ velocities (Gumbel’s was $18.4\times$ too small). They now train the published cross-entropy denoisers with the published conditional fields (the Dirichlet C -factor as in the authors’ code) and decode, as published, the argmax of the final posterior. Neither path

Table S2: Constraint interventions (NFE 100): what each fixes and what it costs.

Alphabet	Arm	Step viol.	x_0 outside	3-mer r
DNA	unconstrained	0.7132	0.9785	0.8943 ± 0.0876
DNA	hard	0.0098	0.9219	0.8961 ± 0.0409
DNA	endpoint	0.0000	0.0022	0.8543 ± 0.0405
Protein	unconstrained	0.9667	1.0000	0.7526 ± 0.0576
Protein	hard	0.0103	0.9980	0.8360 ± 0.0370
Protein	endpoint	0.0100	0.6385	0.5019 ± 0.3607
Codon	unconstrained	1.0000	1.0000	0.1268 ± 0.0233
Codon	hard	0.0159	1.0000	0.1874 ± 0.0396
Codon	endpoint	0.0100	0.7245	0.0714 ± 0.0816

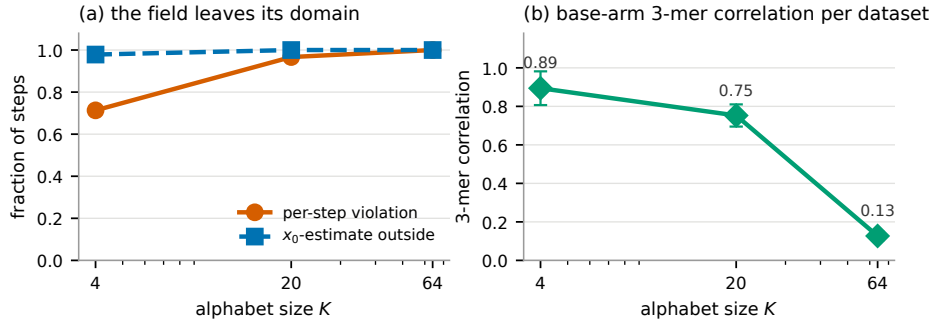


Figure S1: Non-tangency of the base arm against alphabet size (Table 3, base rows).

reaches a vertex, so decoding the final state instead gives lower scores at large K : 3-mer r 0.903 / 0.876 / 0.224 (Gumbel) and 0.904 / 0.911 / 0.375 (Dirichlet). (iii) All Modality 1 numbers were first scored against the training sequences, on data that kept only protein N-termini; they are now scored on held-out rebuilt data. The train-minus-held-out 3-mer gap is small (+0.014 / +0.002 / +0.037 base, +0.011 / +0.002 / +0.050 ours). (iv) Modality 2 first used one feature extractor per run; one shared extractor now scores every arm, and no verdict changed. (v) A guidance rule described as an entropic mirror step was the natural-gradient update, and a stale natural-gradient violation rate was replaced by a rerun. Superseded rows remain in the repository (e.g. training-reference 3-mer r of the first implementations: Dirichlet 0.714 / 0.271 / 0.448, Gumbel 0.586 / 0.499 / 0.041, Fisher 0.967 / 0.908 / 0.421).

A.5 Guidance: Full Grid and Synthetic Check

Table S3 gives every guidance strength on the real sequences. The synthetic check guides a base model on one position with $K = 8$ toward the two rarest symbols, where the ideal guided law (p_{data} restricted to S) is known exactly. Unguided, S is hit with probability 0.0136; every rule reaches 1.000 by $\gamma = 8$ with KL to the ideal law of 0.046–0.066 (Table S4; 100,000 samples). At $\gamma = 8$ the tangent rule escapes with $21\times$ the Euclidean rule’s mass, and the natural gradient cuts the violation rate from 0.97 to 0.77 but needs $4\times$ the strength. At $\gamma = 0$ the rate is already 0.21: the base field, not guidance, leaves the domain.

A.6 Additional Samples and Failure Cases

Diversity (exploratory, added after the results). Marginal KL and 3-mer correlation cannot see a generator that repeats a few sequences, so from the saved NFE-100 samples (4,000 per cell) we measure the unique-sequence fraction and the mean pairwise normalised Hamming distance (Table S5); Hamming distance tracks composition, so the target is the held-out value, not the largest. Base, ours, the ablations, Dirichlet FM and Fisher-Flow match held-out data. The endpoint arms *collapse*: on 3 of 8 protein and 5 of 8 codon seeds (reweighted: 3 and 6) every sample is the same sequence, which is what their low, erratic 3-mer scores reflect. Gumbel-Softmax FM decoded by its posterior is less diverse than the data (Hamming 0.878 vs 0.940 on protein, 0.947 vs 0.982 on codon); decoded by its state it matches (0.940). No arm reproduces a training sequence (largest exact-copy

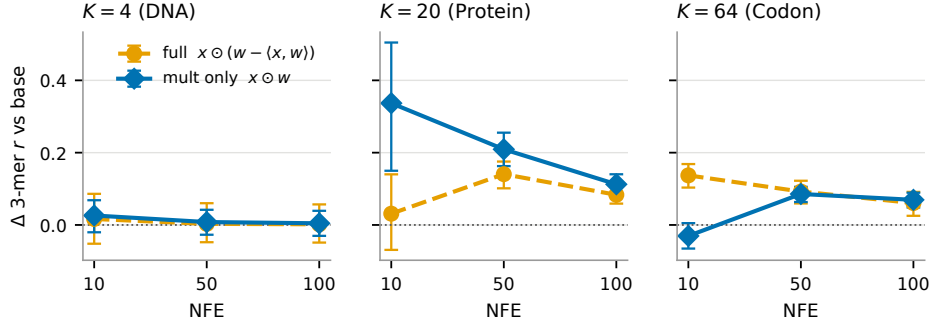


Figure S2: Gain over the base against NFE. Dashed: full replicator. Solid: multiplicative factor alone. At $K=20$ both gains shrink as steps shrink; at $K=64$ the factor's gain appears only from NFE 50.

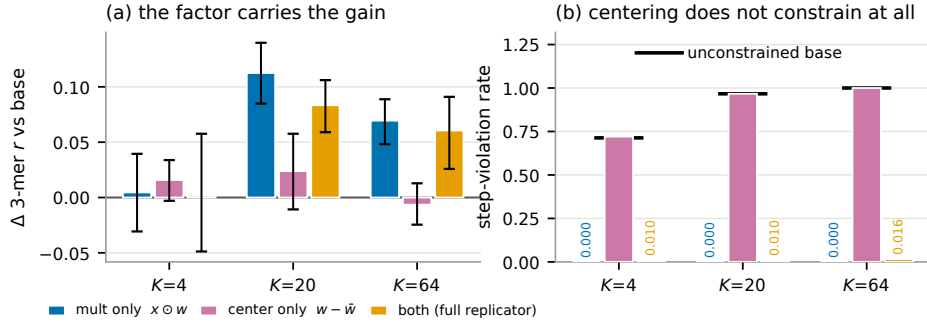


Figure S3: One-at-a-time removal of the two pieces of the replicator form (Table 3).

rate 0.0000). For guidance, where the state is one symbol, the analogue is the split between the two target symbols: the ideal law puts 23% of the target mass on the rarer one; at $\gamma = 8$ the Euclidean and tangent rules put 12% and 12% (under-dispersed), the natural gradient 40% (over-dispersed). None collapses onto one symbol.

Commitment time. The decoded $\arg \max$ of x_t settles early: mean settling times are 0.055, 0.056 and 0.048 for $K = 4, 20, 64$, with 88–96% of positions settled by $t = 0.25$. This is a different quantity from the posterior in Stärk et al. [4, Prop. 1], which is an upper bound, and under a $\text{Dir}(1)$ source much of it follows from the geometry of the linear path rather than from the model. We report it as descriptive only.

A.7 Algorithms

Algorithm 1 Training and sampling with the multiplicative factor (θ is the only trainable component)

```

1: for  $k = 1, \dots, 8000$  do
2:   sample  $x_1 \sim \mathcal{D}$  (one-hot),  $x_0 \sim \text{Dir}(\mathbf{1})$  per position,  $t \sim \mathcal{U}[0, 1]$ 
3:    $x_t \leftarrow (1 - t)x_0 + tx_1$ ;  $v \leftarrow x_t \odot w_\theta(x_t, t)$  ▷ box:  $x_t \odot (1 - x_t) \odot w_\theta$ 
4:    $\theta \leftarrow \theta - \eta \nabla_\theta \|v - (x_1 - x_0)\|_2^2$ 
5: end for
6: Sample:  $x \sim \text{Dir}(\mathbf{1})$ ; for  $i = 0, \dots, N - 1$ :  $x \leftarrow x + \frac{1}{N} x \odot w_\theta(x, \frac{i}{N})$ , clamp at 0, renormalise
7: return  $\arg \max$  per position

```

A.8 Implementation and Reproducibility Details

Every table, figure and in-text number is regenerated from the stored result files by `scripts/make_tables.py` and `scripts/make_figures.py`; the repository README maps each script to the tables it produces and gives setup, data preparation and run commands. PyTorch 2.8 with CUDA 12.8 on the rented GPUs and MPS locally; seeds 0–7; the Gamma sampler used by the Dirichlet baseline was validated against `scipy` by Kolmogorov–Smirnov test

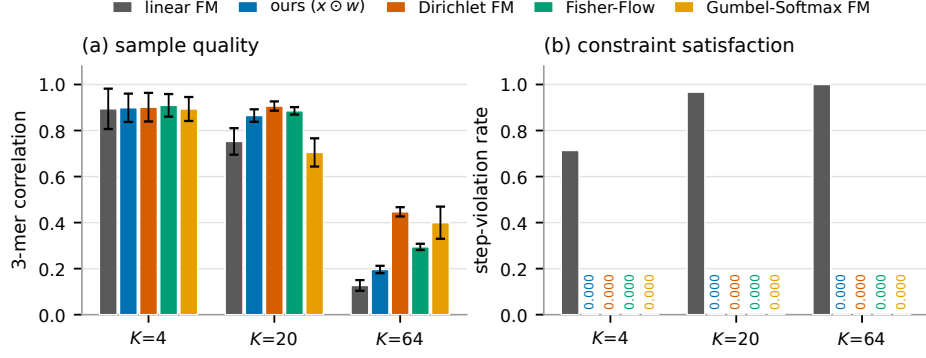
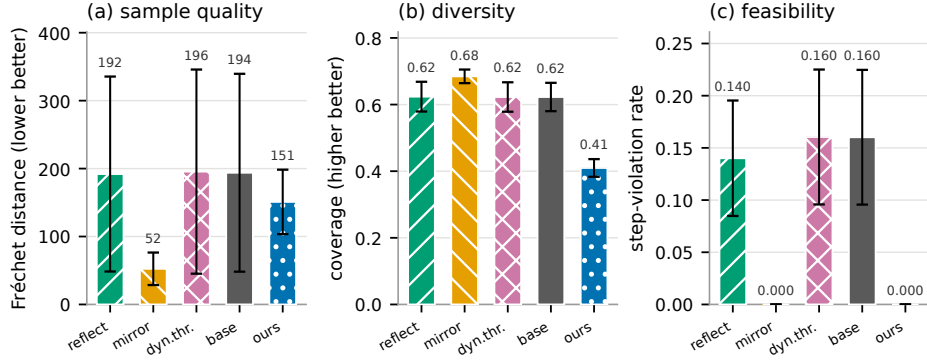


Figure S4: Matched comparison with recent methods (Table 4 in graphical form).



MNIST, $[0, 1]^{784}$, U-Net backbone, NFE = 100, 8 seeds. Feature extractor test accuracy 96.2%.
External arms are our re-implementations. Bars are means; whiskers are ± 1 s.d.

Figure S5: Modality 2 transfer (Table 5 in graphical form).

($p = 0.589/0.441/0.629$ at shapes 1.0/2.5/9.0). Baselines are re-implemented from their papers; the Dirichlet C -factor follows the authors' reference code and matches it to 10^{-10} .

Table S3: Real-data guidance, full γ grid (8 seeds).

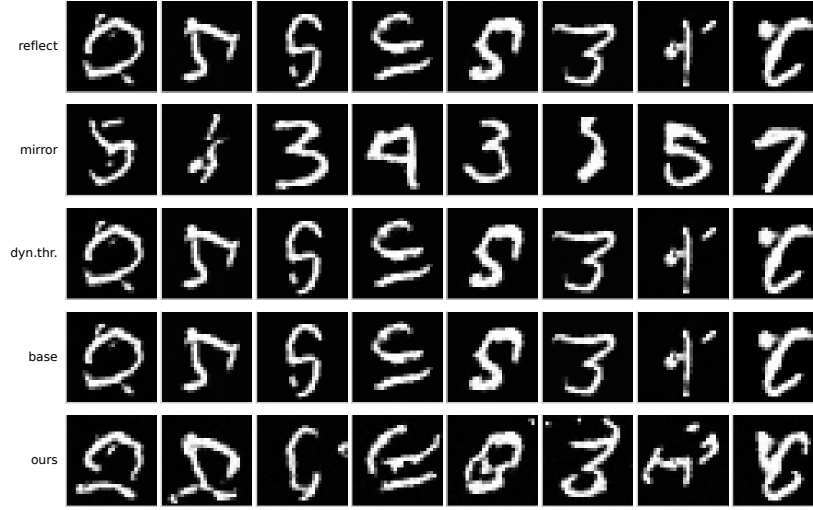
Property	Guidance (γ)	Target frac. $\uparrow$	Hit rate $\uparrow$	3-mer r held-out	3-mer r top decile $\uparrow$	Hamming (diversity)
DNA (GC)	0	0.510 ± 0.026	0.09 ± 0.05	0.901 ± 0.058	0.588 ± 0.156	0.748 ± 0.002
	0.01	0.537 ± 0.025	0.14 ± 0.07	0.859 ± 0.073	0.749 ± 0.104	0.746 ± 0.003
	0.03	0.589 ± 0.023	0.30 ± 0.10	0.715 ± 0.071	0.916 ± 0.041	0.740 ± 0.005
	0.1	0.742 ± 0.018	0.93 ± 0.03	0.368 ± 0.033	0.939 ± 0.029	0.689 ± 0.009
	0.3	0.961 ± 0.005	1.00 ± 0.00	0.040 ± 0.011	0.649 ± 0.020	0.533 ± 0.008
	1	1.000 ± 0.000	1.00 ± 0.00	0.038 ± 0.023	0.579 ± 0.058	0.469 ± 0.051
	3	1.000 ± 0.000	1.00 ± 0.00	0.043 ± 0.028	0.553 ± 0.068	0.438 ± 0.069
	<i>held-out real</i>	0.495	0.09	—	—	—
Protein (hydrophobic)	0	0.356 ± 0.011	0.03 ± 0.01	0.855 ± 0.039	0.598 ± 0.054	0.943 ± 0.001
	0.01	0.390 ± 0.014	0.12 ± 0.05	0.823 ± 0.062	0.675 ± 0.058	0.943 ± 0.001
	0.03	0.455 ± 0.017	0.55 ± 0.13	0.697 ± 0.084	0.742 ± 0.067	0.941 ± 0.001
	0.1	0.639 ± 0.021	1.00 ± 0.00	0.383 ± 0.061	0.683 ± 0.058	0.921 ± 0.004
	0.3	0.951 ± 0.015	1.00 ± 0.00	0.217 ± 0.020	0.532 ± 0.022	0.841 ± 0.011
	1	1.000 ± 0.000	1.00 ± 0.00	0.275 ± 0.015	0.531 ± 0.035	0.747 ± 0.037
	3	1.000 ± 0.000	1.00 ± 0.00	0.269 ± 0.022	0.495 ± 0.054	0.715 ± 0.057
	<i>held-out real</i>	0.351	0.10	—	—	—

Table S4: Synthetic guidance check, $K=8$ target (one seed). Full sweep: Table S6.

Rule	$\gamma = 1$			$\gamma = 8$		
	Hit $\uparrow$	KL $\downarrow$	Viol. $\downarrow$	Hit $\uparrow$	KL $\downarrow$	Viol. $\downarrow$
Unguided ($\gamma=0$)	0.0136	4.30	0.210	—	—	—
euclidean	0.9994	0.071	0.884	1.0000	0.048	0.972
tangent	0.9995	0.070	0.977	1.0000	0.046	1.000
natural	0.4811	0.735	0.267	1.0000	0.066	0.766

Table S5: Exploratory diversity, Modality 1 (8 seeds, 4,000 samples, NFE 100; mean $\pm$ s.d.). Denoiser baselines decoded by their posterior, as reported elsewhere. † A sample-saving rerun of the §4.2 families; it reproduces the registered marginal KL exactly.

Arm	Pairwise Hamming (NFE 100)			Unique fraction		
	DNA	Protein	Codon	DNA	Protein	Codon
<i>Held-out real</i>	0.750 ± 0.000	0.940 ± 0.000	0.982 ± 0.000	1.00	1.00	1.00
<i>Unigram (no model)</i>	0.750 ± 0.000	0.941 ± 0.000	0.982 ± 0.000	1.00	1.00	1.00
Linear FM (base)	0.747 ± 0.003	0.944 ± 0.001	0.980 ± 0.002	1.00	1.00	1.00
Ours $x \odot w$	0.748 ± 0.002	0.944 ± 0.001	0.982 ± 0.001	1.00	1.00	1.00
Center-only	0.748 ± 0.003	0.943 ± 0.001	0.980 ± 0.003	1.00	1.00	1.00
Full replicator	0.748 ± 0.001	0.944 ± 0.001	0.982 ± 0.001	1.00	1.00	1.00
Endpoint	0.749 ± 0.000	0.588 ± 0.455	0.368 ± 0.475	1.00	0.63	0.38
Endpoint, reweighted	0.749 ± 0.001	0.587 ± 0.454	0.245 ± 0.424	1.00	0.63	0.25
Dirichlet FM	0.748 ± 0.002	0.939 ± 0.002	0.981 ± 0.001	1.00	1.00	1.00
Fisher-Flow	0.748 ± 0.001	0.941 ± 0.002	0.981 ± 0.001	1.00	1.00	1.00
Gumbel-Softmax FM	0.746 ± 0.003	0.878 ± 0.011	0.947 ± 0.004	1.00	1.00	1.00
Flow matching, NFE $10^\dagger$	0.747 ± 0.002	0.911 ± 0.023	0.872 ± 0.081	1.00	1.00	1.00
Diffusion (DDIM) †	0.687 ± 0.042	0.896 ± 0.049	0.941 ± 0.034	1.00	1.00	1.00
Diffusion, NFE $10^\dagger$	0.464 ± 0.157	0.602 ± 0.136	0.786 ± 0.217	1.00	1.00	1.00



First eight samples drawn at NFE=100, seed 0, saved before any inspection. The clamping-family arms are pixel-identical.

Figure S6: Uncurated Modality 2 samples (the first eight drawn at NFE 100, seed 0, saved before inspection). Our arm attains a lower FD than the base on this seed yet is visibly blotchier: FD in a small classifier’s features is not a perceptual judgement.

Table S6: Full guidance sweep; Table S4 reports the $\gamma \in \{1, 8\}$ columns.

Rule	γ	Hit rate $\uparrow$	KL to ideal $\downarrow$	Step viol. $\downarrow$
euclidean	0.0	0.0136	4.2978	0.2102
euclidean	0.5	0.5642	0.6748	0.7134
euclidean	1.0	0.9994	0.0710	0.8837
euclidean	2.0	1.0000	0.0646	0.9293
euclidean	4.0	1.0000	0.0513	0.9541
euclidean	8.0	1.0000	0.0478	0.9723
tangent	0.0	0.0140	4.2695	0.2106
tangent	0.5	0.5674	0.6728	0.9479
tangent	1.0	0.9995	0.0698	0.9772
tangent	2.0	1.0000	0.0655	0.9921
tangent	4.0	1.0000	0.0514	0.9982
tangent	8.0	1.0000	0.0461	0.9999
natural	0.0	0.0138	4.2810	0.2104
natural	0.5	0.1264	2.0826	0.1691
natural	1.0	0.4811	0.7355	0.2674
natural	2.0	0.9430	0.1072	0.4950
natural	4.0	1.0000	0.0588	0.6660
natural	8.0	1.0000	0.0665	0.7662